Automated Biometric System for Individual Re-Identification of Mugger Crocodile

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***Abstract*—Individual identification of free-ranging wildlife is a key concern for ecological monitoring and conservation, however, conventional invasive tagging methods pose significant risk to the target species. We aim to create an automated system for individual re-identification of mugger crocodiles (Crocodylus palustris) using deep learning on UAV-captured dorsal imagery. In this work, two models have been implemented and compared — YOLOv5l with bounding box annotation and Inception-v3 without annotation. We follow an open-set evaluation protocol using two test sets measuring True Positive Rate (TPR) for known individuals and True Negative Rate (TNR) for unknown ones, with a fixed decision threshold of 0.84. On known subjects, YOLOv5l outperformed Inception-v3 by having a higher TPR of 100% and on unknown rejection, Inception-v3 is better with a TNR of 96.7 on Test-II and 96.9 on Test-III, indicating that bounding box elimination can enhance re-identification but does not always enhance the rejection of unseen individuals.**

***Index Terms*—Mugger Crocodile, Individual re-identification, YOLOv5l, Inception-v3, deep learning, Open-set recognition**

I. Introduction

The technique of monitoring the wildlife populations needs to be aware of which individual animal is being tracked at a particular time. Such monitoring is urgently required in the case of the mugger crocodile (Crocodylus palustris), a freshwater crocodile which is Vulnerable by the IUCN.

In recent years, the process of automated animal identification has undergone a revolution through the application of deep learning and convolutional neural networks (CNNs). The use of CNNs has been successful to identify members of other species, including tigers using stripe patterns, chimpanzees using facial features, and small birds using patterns of feathers. These techniques offer less invasive alternative compared to traditional tagging. One of the other architectures examined, YOLO-based [3] models with bounding box annotations, have been especially promising because these architecture are able to look at the specific part of the body that is relevant and ignore noisy background information.

The mugger crocodile has provided a special chance to the CNN-based individual identification since every individual holds a different dorsal scute pattern; hard calcium bony plates, in a unique arrangement upon the back body. These scute patterns are a natural biometric identification design, just like in humans, fingerprints. Data collection through UAV has enabled high-quality dorsal images of free-ranging crocodiles to be taken without disturbing them that offers a scalable method of constructing large training datasets of wild populations across geographically varied locations.

One of the most crucial issues in the real-life implementation of such a system is the open-set recognition problem. When conducting a field study, a researcher will come across people who have been catalogued before as well as totally new and unnoticed crocodiles at the same time. Closed-set classifier, which never makes an error in classification, would falsely label all unknown crocodiles as one of the trained ones. The system should hence not only be able to re-identify known people but also reliably identify unknown ones. That needs to be done with a careful threshold-based decision making.

In this work we use and compare two deep learning baselines of mugger crocodile re-identification in this paper. The former is YOLOv5l, which takes into consideration annotated bounding boxes and concentrates on the back part and carries out classification. The second is Inception-v3 [4] and takes the entire image as an input and is more influenced by the background noise. Both models are trained using a set of 4,094 UAV images of 6 people and tested on an open-set protocol with the threshold of 0.84. True Positive Rate (TPR) is used to measure performance with known individuals and True Negative Rate (TNR) is used to measure performance with unknown individuals as per [1].

II. Methodology

A. Dataset

The dataset used in this work is a subset of the UAV-captured dorsal images collected by [2] from free-ranging mugger crocodiles in Gujarat, India. While the original study consisted of 143 individual classes across 19 locations, we focus on a narrower subset of 6 individual classes, referred to by their class labels: 107, 108, 109, 110, 111, and 112. The total dataset comprises 4,094 images captured using a drone at a height of 8–10 meters above the crocodile, focusing on the dorsal scute region. Images were captured during morning basking hours between 0900 and 1200 h, under consistent lighting conditions, at a resolution of 3840×2160 pixels.

B. Annotation Pipeline

Since manual annotation of all 4,094 images was not feasible, we developed a two-stage annotation pipeline. In the first stage, we manually annotated 300 images using Label Studio, where bounding boxes were drawn around each crocodile's dorsal region. Class IDs were mapped as 107→0, 108→1, 109→2, 110→3, 111→4, and 112→5.

In the second stage, a single-class YOLOv5l detector was trained on these 300 manually annotated images, treating all individuals as one generic class ”crocodile”. We run this detector on the remaining 3,794 unannotated images to generate predicted bounding boxes. The class ID for each auto-annotated image is taken from its filename rather than the model output, preserving individual identity information in the labeled files. At the end of phase 1 of the project, after completing this pipeline, all 4,094 images had corresponding YOLO-format annotation files.

C. YOLOv5l Model

The YOLOv5l model was selected as the primary model due to its strong performance in object detection and classification tasks. The model was trained on all 4,094 annotated images across 6 classes. The dataset was split into 90% training and 10% validation using a stratified split, ensuring each individual was proportionally represented in both sets. Training hyperparameters follow [1]: 20 epochs, batch size 5, SGD optimizer with a learning rate of 0.01, and binary cross-entropy with logits loss. Input image size was set to 640×640.

At the inference time, a two-threshold approach was applied. We applied a detection confidence threshold of 0.25 to draw a bounding box around any detected crocodile, irrespective of its identity. We applied another separate identity threshold of 0.84, the Equal Error Rate (EER) threshold inspired from the [1], to the class confidence score. If the confidence exceeded 0.84, the mugger crocodile was assigned a known identity label. If it fell below this threshold, the detection was labelled as ”unknown.” This design allows the model to operate effectively in open-set conditions where unseen mugger crocodile individuals may appear.

D. Inception-v3 Model

The Inception-v3 model was considered as the second model to allow a direct comparison with a non-bounding-box approach. Inception-v3 takes the full image as input without any prior cropping or bounding box-based region of interest extraction, unlike what YOLOv5l did. Images were resized to 224×224 pixels as required by the architecture. The same 6-class setup and train-validation split were used. Training was performed using RMSprop optimizer, learning rate 0.01, batch size 5, categorical cross-entropy loss, and 20 epochs, consistent with the [1]'s hyperparameter settings.

Since Inception-v3 does not use bounding boxes, the entire image — including background elements such as water, mud, and surrounding vegetation — is passed to the network during both training and inference. This introduces background noise into the feature learning process, which negatively impacts classification accuracy compared to the bounding box-guided YOLOv5l approach.

III. Results and Discussions

Both models were evaluated on three test sets. Test-I and Test-II are inspired from the open-set dataset provided by [2]. Test-I contains images of the 6 known trained individuals mixed with unknown muggers, used to measure TPR. Test-II contains only unknown muggers and non-mugger species (Gharial and Saltwater crocodile), used to measure TNR. Test-III is an additional set of 32 images of 16 unknown individuals provided externally, containing no known individuals, used purely to evaluate TNR.

TABLE I
Performance Comparison of YOLOv5l and Inception-v3 Across Test Sets

Test Set	Model	Known	Unknown	TPR	TNR	Accuracy
Test-I	YOLOv5l	12	228	100.0%	85.09%	92.54%
Test-I	Inception-v3	12	228	83.33%	94.74%	89.04%
Test-II	YOLOv5l	0	240	N/A	88.75%	–
Test-II	Inception-v3	0	240	N/A	96.67%	–
Test-III	YOLOv5l	0	32	N/A	40.62%	–
Test-III	Inception-v3	0	32	N/A	96.88%	–
Paper (ref.)	–	–	–	88.8%	89.6%	89.2%

On Test-I, achieved a perfect TPR of 100%, correctly re-identifying all 12 known individual images, while Inception-v3 achieved 83.33%, missing 2 out of 12. However, Inception-v3 showed a higher TNR of 94.74% compared to YOLOv5l's 85.09%, meaning it was more conservative in assigning known identities to unknown crocodiles.

On Test-II, which contained only unknown muggers and non-mugger species, both models performed well in rejecting the unknown and non-mugger species. Inception-v3 achieved a TNR of 96.67% versus YOLOv5l's 88.75%. Notably, both models correctly rejected 100% of Gharial and Saltwater crocodile images, consistent with the [1]'s finding of 100% TNR for non-mugger species.

On Test-III, the externally provided set of 32 unknown images, Inception-v3 maintained a strong TNR of 96.88%, while YOLOv5l performed significantly worse at 40.62%. Most false positives from YOLOv5l on this set were incorrectly assigned to class 109, suggesting the model may have overfit certain visual features of that individual.

IV. Conclusion

In this work, we present an automated system for individual re-identification of mugger crocodiles using deep learning on UAV imagery. A scalable annotation pipeline was developed using YOLO-based auto-annotation.

Experimental results show that YOLOv5l achieves superior re-identification performance with a TPR of 100%, while Inception-v3 demonstrates stronger unknown rejection with higher TNR across test sets. This highlights a trade-off between accurate identification and open-set generalization. Future work will focus on improving open-set decision mechanisms and combining the strengths of both approaches for more reliable real-world deployment.

References

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